

Multivariate Correlation Self-Distillation Transformer for Time Series Forecasting With Incomplete Data

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Abstract—Multivariate time series forecasting estimates future development by capturing variable relationships and constructing temporal regular, which is widely used in many scenarios, including industrial production, economic development, and disease prediction. Although the existing deep learning methods have achieved impressive results in multivariate time series forecasting tasks, the existing methods only emphasize the prediction performance and ignore the widespread issue of missing data in the real world. This article proposes a robust multivariate correlation self-distillation Transformer framework for incomplete time series data forecasting. The proposed method first decouples the interinference of historical series and the exter-inference of future series into two stages. The first stage focuses on the reconstruction of historical series, while the second stage focuses on the prediction of future series. Then, a novel multivariate correlation Transformer is designed as the basic component of the network, which can perform feature inference from both multivariate relationships and single-variate temporal regular. Finally, a variable correlation self-distillation method is proposed to self-distill the more complete variable relationship from the exter-inference stage to the interinference stage. The

proposed method is verified on eight real-world datasets, and both qualitative and quantitative results show that the proposed method has good performance.

Index Terms—Incomplete time series data, multivariate time series forecasting, self-distillation, transformer.

I. INTRODUCTION

MULTIVARIATE time series forecasting tasks aim to extract the relationship between variables, mine the temporal feature regular, and finally predict the future development trend [1]. Multivariate time series forecasting has a wide range of real-world applications, such as industrial capacity soft measurement [2], weather forecasting [3], and disease spread estimation [4]. Benefitting from the excellent complex time feature modeling ability of deep networks, multivariate time series forecasting based on deep learning technology has become the mainstream. Early research focused on the construction of long time features, and the classical gated network techniques are developed. At present, there are various researches on multivariate time series forecasting, including Transformer-based methods [5], frequency domain methods [6], patch-based methods [7], etc. These methods have achieved impressive performance on many real-world multivariate time series forecasting tasks.

However, there is missing data in many practical scenarios. The reasons for missing data are varied, for example, sensor failures in industrial scenarios [8] and signal transmission errors in traffic scenarios [9]. If the deep learning model only uses the complete time series data during training or just uses simple data preprocessing to compensate for missing data (such as interpolation and zero-value padding), the well-trained models will lead to model deviation, sudden changes in neuron responses, and significant performance decreases during actual deployment and testing.

Some existing studies attempt to estimate missing data from available data by designing learnable models, and this category of studies is called time series imputation tasks [10]. The most common time series imputation schemes are based on recurrent neural networks. This category of methods is close to the time series forecasting, where recurrent neural networks are used to construct the time series features of the available data and then estimate the missing series from the available data features.

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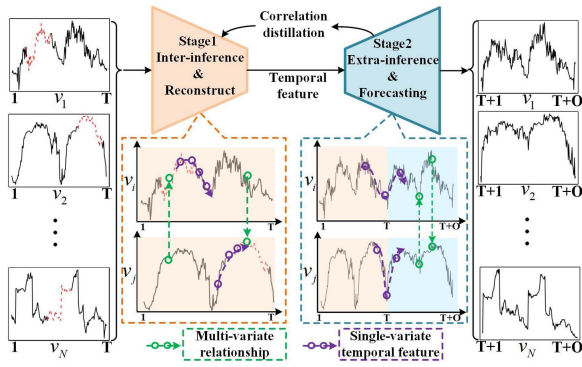


Fig. 1. Multivariate correlation Transformer self-distillation method.

Yoon et al. [11] proposed a multidirectional recurrent neural network, which uses temporal features to perform interpolation within data streams and across data streams. At present, there are few studies that develop integrated schemes for time series forecasting with missing data. Tran et al. [12] proposed an end-to-end time series imputation and forecasting model, which only uses auxiliary variables to predict the target variable, and does not focus on multivariate data. The existing time series imputation studies also only focus on the reconstruction of missing data, without integrating the time series forecasting task or regarding time series forecasting as an independent downstream task [13]. Obviously, missing data reconstruction and future time series forecasting both need to extract and infer the temporal features in the observed series, and the two tasks are promising to be integrated into a unified framework.

In this work, we propose a multivariate correlation self-distillation Transformer framework for time series forecasting with missing data. Our proposal is presented in Fig. 1. Common forecasting models usually include historical time series feature extraction and future time series mapping, but there is no clear distinction between these two processes [14]. Since the proposed method aims to solve the time series forecasting of incomplete data, we clearly divide the overall framework into two stages: internal inference for historical time series and external inference for future time series.

The first stage models the historical time series features and uses a reconstruction loss to constrain the reconstructed missing data. The second stage uses the reconstructed historical time series features as input to predict the future time series. Effective information in multivariate time series forecasting mainly includes two parts. One is the complementary information between variables with positive correlation (green dots in Fig. 1), and the other is the temporal regular within the single-variate (purple dots in Fig. 1). In order to fully exploit these two kinds of information, we design a multivariate correlation Transformer (MVCT) as the basic component of the overall framework. The Transformer includes a novel cosine similarity attention mechanism and a single-variate feedforward network (FFN). The cosine similarity attention mechanism can capture similar variables, and the single-variate FFN only extracts the temporal features inside the single-variate. We also propose a variable correlation self-distillation method for optimizing the two-stage framework. Compared with the features in the first stage, the

features in the second stage are more complete and contain the predicted future series features, so the variable relationship in the second stage can better reflect the real variable relationship. The proposed variable correlation self-distillation distills the multivariate correlation from the second stage to the first stage. The contributions of the proposed method can be summarized as follows.

- 1) We propose a two-stage framework for inter-inference and reconstruction of historical time series, and external inference and prediction of future time series.
- 2) A novel MVCT is designed, in which the cosine similarity attention and the single-variate FFN extract the correlation information between variables and the single-variate temporal regular, respectively.
- 3) A variable correlation self-distillation scheme is proposed to backward distil the more complete variable relationship to the first stage of the network.
- 4) Experiment results on eight real multivariate datasets show that the proposed method is superior to state-of-the-art methods. It has a good forecasting effect even when the data missing rate reaches 50%.

II. RELATED WORK

A. Time Series Imputation

Time series imputation can reconstruct the missing time series. The method based on time feature is the most common, which captures the time regularity and predicts the missing series. Che et al. [15] proposed a gated recurrent unit to integrate different types of data missing patterns into a deep network, which makes the network robust to missing data. Some studies also attempted to use information from other dimensions to compensate missing data. Ma et al. [16] proposed a cross-dimensional self-attention mechanism, which integrates time, location, and measurement information for time series imputation. Du et al. [17] designed a diagonal mask self-attention module, which can explicitly capture the time dependence and variable correlation, and can dynamically assign learning weights according to missing information. With the development of generative technology, time series data imputation based on generative adversarial networks has also been developed [18]. Luo et al. [19] proposed to use a modified gated recurrent network to model the temporal irregularity and act as a generator network. Then, it is jointly trained with the discriminator network. Fortuin et al. [20] proposed a variational autoencoder (VAE) method with structured variational approximation, which represents the time series as a low-dimensional representation and then performs the reconstruction more smoothly. It should be noted that most of the current time series imputation methods only focus on the reconstruction within the historical series, and the proposed method in this article reconstructs the missing data while completing the future time prediction.

B. Time Series Forecasting

There has been a lot of researches on multivariate time series forecasting based on deep learning. Early research mainly focused on recurrent neural networks and 1-D convolutional

networks. Zhao et al. [21] proposed an improved LSTM model and constructed a 2-D network using multiple memory units to obtain the spatio-temporal correlation of the transportation system. Oreshkin et al. [22] developed a lightweight temporal convolution network to enhance local information about weather changes. In addition to explicit time regular, there are also implicit complex time patterns, such as periodic fluctuations, long-term trends, and short-term trends. Methods based on time series decomposition are expected to mine implicit time patterns. The most classical time series decomposition method is period-trend decomposition. Wang et al. [23] proposed a multiscale isometric convolution, which constructs multiscale period and trend terms by changing the size of the decomposition kernel, and then predicts the future period and trend terms, respectively. Based on the multiscale period term and trend term, the TimeMixer [24] method integrates the periodic term bottom-up and integrates the trend term top-down.

Although these methods have a certain ability to capture the time trend, they are limited by the receptive field and cannot construct long-term dependence. Transformer-based methods can solve this problem well. Transformer takes the multihead attention mechanism as the core [25]. Transformer models can be divided into two categories. The first focuses on learning long-term time dependencies from the time dimension. Wu et al. [5] proposed autoformer, which combines time series decomposition, and uses a periodic autocorrelation to perform correlation aggregation from the subseries level. Kitaev et al. [26] proposed an efficient Transformer that replaces the attention mechanism with locality-sensitive hashing dot product attention. The second type of Transformer method focuses on the correlation between multiple variables, and uses the attention mechanism to capture the similar variables to improve the prediction effect. Zhang et al. [27] proposed crossformer, which contains dimension segment wise and two stage attention. Dimension segment wise can transform temporal information and variable information into a 2-D array. Then, two stage attention strengthens the two. Liu et al. [28] believed that many variables have similar temporal features, and designed inverted transformers, which embed tokens in the variable dimension and use attention to extract variable correlation. The complementary information between similar variables can also make up for the missing data information, so this article proposes a novel MVCT based on cosine similarity as the basic component.

C. Knowledge Distillation

Knowledge distillation technology is usually used for model compression. The base model contains a large teacher network with good learning ability and a smaller student network, and the knowledge in the teacher network is transferred to the student network by designing an appropriate distillation loss. In the time series task, Liu et al. [29] proposed a method combining knowledge distillation and graph neural network, it extracts electricity load knowledge from the public data base and distill it into the individual household electricity load forecasting model. Xu et al. [30] designed a contrastive adversarial knowledge distillation scheme to improve fine-grained feature alignment

in time series regression task, it maximizes the similarity of the same sample between the teacher model and the student model. Liu et al. [31] proposed a cross-modal knowledge distillation method to align large-scale language models with time series prediction models and improve the generalization ability of prediction.

The self-distillation method considers that different stages or layers of a network contain different knowledge, and knowledge distillation can be performed in the same network [32]. In the proposed method, the second stage of the overall framework (future time series prediction) contains longer and more complete time series features, which better reflects the real relationship between variables. We distill the variable relations of the second stage to the first stage by self-distillation.

III. METHODOLOGY

A. Problem Definition

The multivariate time series forecasting task is given a historical series $\tilde{X}$ with T time steps, to predict the future series $\tilde{Y}$ with O time steps. $\tilde{X} = \{v_{1:T}^1, \dots, v_{1:T}^N\} \in \mathbb{R}^{N \times T}$, $\tilde{Y} = \{v_{T+1:T+O}^1, \dots, v_{T+1:T+O}^N\} \in \mathbb{R}^{N \times O}$, v is the time series variable, and N is the number of variables. Here, a binary matrix M is used to represent the location of the missing value, $M \in \mathbb{R}^{N \times T}$, in which 1 represents the available value and 0 represents the missing value. Historical series with missing data can be represented using $X \in \mathbb{R}^{N \times T}$ and $X = M\tilde{X}$. The goal of the proposed method is to build a two-stage time series forecasting model

$$\begin{aligned} X_{\text{Rect}} &= f_1(X) \\ Y_{\text{Pred}} &= f_2(X, f_1(X), M) \end{aligned} \quad (1)$$

where $f_1(\cdot)$ models the internal temporal features of the historical series with missing data from time 1 to time T and reconstructs the missing part. $f_2(\cdot)$ extrapolates the historical series features outward to obtain the predicted series between time $(T + 1)$ and time $(T + O)$.

B. Overall Framework

The overall proposed framework is presented in Fig. 2 and contains two stages: interinference network and exter-inference network. We try to build a prediction method that is robust to any missing data, so the missing matrix M randomly selects the missing positions according to a certain missing rate, which is set to a random value between $[0, 0.5]$ during training. The series data with missing values are fed into the inter-inference network to model and reconstruct temporal features of the series from time 1 to time T , the inter-inference network contains a feature embedding module and several MVCTs. Feature embedding is performed first, the input feature is $X \in \mathbb{R}^{N \times T}$. Timestamp $[1, \dots, T]$ is a matrix of dimension $\mathbb{R}^{4 \times T}$, where the 4 in the first dimension represents the year, month, day, and hour information. Then, X and $[1, \dots, T]$ are connected in the first dimension and then add the position encoding Pos to form a matrix of dimension $(N + 4) \times T$. The first dimension is compressed to N by a Linear layer to achieve the embedding of time features

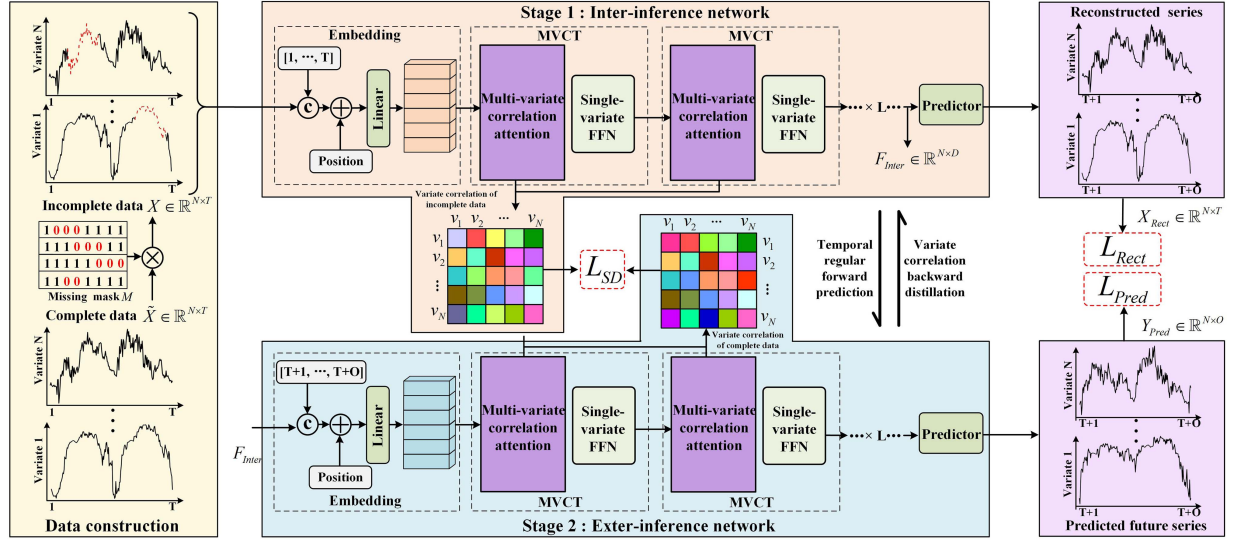


Fig. 2. Overall framework of the proposed multivariate correlation self-distillation Transformer, which mainly includes data construction, inter-inference network, exter-inference network, and self-distillation part.

into variable features. The second dimension is then mapped to D also by a linear layer for subsequent Transformer operations, D is set to 128. The feature embedding can be expressed as follows:

$$\begin{aligned} (X \odot [1, \dots, T]) + \text{Pos} &\rightarrow X \in \mathbb{R}^{(N+4) \times T} \\ \text{Linear}(X) &\rightarrow X \in \mathbb{R}^{N \times T} \\ \text{Linear}(X) &\rightarrow F \in \mathbb{R}^{N \times D} \end{aligned} \quad (2)$$

where $\odot$ denotes concatenation operation, F is the embedding feature, D is the dimension of the embedding feature, and $\text{Linear}(\cdot)$ is a linear layer. Position encoding Pos is generated using the common sine and cosine position encoding functions.

Then, several MVCT layers infer the feature F from multivariate complementary information and single-variate temporal regular, and obtain the reconstructed feature F_{Inter} of the historical series. F_{Inter} is processed by a linear layer predictor to output the reconstructed history series, and an auxiliary loss L_{Rect} is used to constrain. The exter-inference network also contains a feature embedding module and multiple MVCT layers, and takes the reconstructed F_{Inter} as input. Since the exter-inference network needs to map the historical series features into the future series, the feature embedding model first uses a linear layer to adjust the dimension of the future series timestamp $[T+1, \dots, T+O]$ to $4 \times D$ and then does the embedding. Finally, the features of the exter-inference network are directly mapped to the future series, which is constrained by the predict loss function L_{Pred} .

C. Multivariate Correlation Transformer

The MVCT is the basic component of the overall network and is shown in Fig. 3. It mainly includes a novel multivariate cosine similarity attention mechanism and a single-variate FFN. The MVCT constructs temporal regularity between the related variables. Temporal regularities are complemented between variables with similar temporal regularities. Meanwhile, the temporal regularity of the single-variate is considered for both

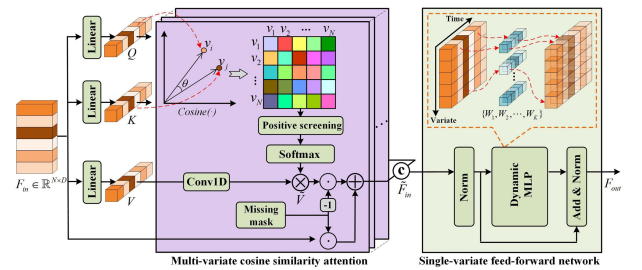


Fig. 3. Proposed MVCT, which mainly includes multivariate cosine similarity attention and single-variate FFN.

related and unrelated variables. Both types of information are important for internal inference of historical series and external inference of future series, so MVCT can adapt to complex time series. Specifically, given a feature $F_{\text{in}} \in \mathbb{R}^{N \times D}$ as the input to MVCT. We first use a linear layer to map it into query vector (Q), key vector (K), and value vector (V)

$$\begin{aligned} Q, K, V &= \text{Linear}(F_{\text{in}}) \\ Q, K, V &\in \mathbb{R}^{N \times D}. \end{aligned} \quad (3)$$

The traditional attention in Transformer uses the dot product method to obtain the correlation between the Q and the K , but the multivariate time series take variable as token and each variable has missing information in the time dimension. Therefore, using the dot product method to obtain the correlation of variables will magnify the impact of these local missing information and weaken the overall relationship of variables. We design a novel multivariate cosine similarity attention for time series with missing data. First, we compute the correlation between Q and K using the cosine similarity function [33]

$$S(i, j) = \frac{\langle Q(:, i), K(:, j) \rangle}{\|Q(:, i)\| \|K(:, j)\|} \quad (4)$$

where $\langle \cdot, \cdot \rangle$ is the inner product computation, $\|\cdot\|$ is the L2 normalization, and $S(i, j)$ represents the correlation between

the i th and j th variables, $S \in \mathbb{R}^{N \times N}$. The variables that have a positive correlation in the cosine similarity have a correlation greater than 0, and we screen the positively correlated variables

$$S(i, j) = \begin{cases} S(i, j), & \text{if } S(i, j) > 0 \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Then, a Softmax function is used to get the variable correlation score $\tilde{S}$

$$\tilde{S}(i, j) = \frac{\exp(S(i, j))}{\sum_{k=1}^N \exp(S(i, k))}. \quad (6)$$

Finally, the V vector is weighted by the correlation score. In order to maximize the complementarity between variables, a 1-D convolution with N kernels is used to process V

$$\tilde{V} = \text{Conv1d}(V) \tilde{S}. \quad (7)$$

The inter-inference network focuses on missing series reconstruction. $\tilde{V}$ is a feature with variable complementary information built by cosine similarity attention. F_{in} can be regarded as a original feature of the original time series data. By observing the correspondence between the missing information and the feature response of the F_{in} , we find that F_{in} still retains most of the original data characteristics, such as the location of missing values. Therefore, we use $(1 - \tilde{M})$ to focus on the missing features in the complementary feature and then add them to F_{in} to complete the reconstruction of missing information. The dimension of M is changed by interpolation (resize operation) to $\tilde{M} \in \mathbb{R}^{N \times D}$. This process is written as

$$\tilde{F}_{\text{in}} = \tilde{V} \odot (1 - \tilde{M}) + F_{\text{in}} \odot \tilde{M} \quad (8)$$

where $\odot$ is the matrix elementwise multiplication and $\tilde{F}_{\text{in}}$ is the reconstructed features. $\tilde{F}_{\text{in}}$ will also participate in subsequent single-variate FFN and Transformer calculations, so as long as most of the reconstructed features can participate in the subsequent calculations, it will have a better effect than using only the missing features. It should be noted that the exter-inference stage maps features to future time series, so this operation is not included.

After that the features go through the single-variate FFN. Convolution layer is usually used in traditional FFN, in which the number of convolution kernels is the same as the number of variables, which means that convolutional kernel sliding on the time dimension will mix all the variable information. Here, we design a dynamic multilayer perceptron (MLP) to perform single-variate inference. The process can be written as follows:

$$\tilde{F}_{\text{in}} = \sum_{k=1}^K \delta_k(\tilde{F}_{\text{in}}) W_k \tilde{F}_{\text{in}}. \quad (9)$$

The traditional single kernel layer may have potential bias in the process of neural network learning, and incomplete time series will further lead to inaccurate and biased temporal regularity. The traditional layer may fall into this specific bias in the learning process. The dynamic MLP contains multiple perceptrons, which can be dynamically adjusted according to the input data. It can be regarded as a small multiexpert system to avoid the errors made by a single expert. The dynamic MLP first computes the input using multiple kernels and then computes the weight of

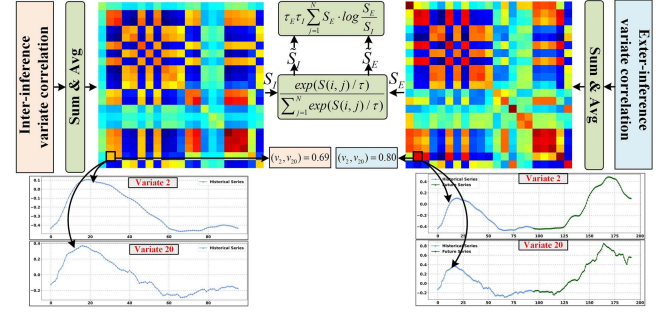


Fig. 4. Calculation process of the variable correlation self-distillation, here, we use a case of Weather dataset. It can be seen that variate 2 and variate 20 have a very high similarity in this case, and the exter-inference network can capture the relationship between the two variables more accurately.

each kernel based on the input. $\{W_1, W_2, \dots, W_k\}$ are the parameters of the perceptrons, and the dimension of each parameter is $D \times D$, so that the information of the input variable is only passed to the corresponding output variable after the parameter mapping, which ensures the inference of the single-variate. We set K to 3. Meanwhile, $\delta(\tilde{F}_{\text{in}}) = \text{Softmax}(W_\delta \text{Avg}(\tilde{F}_{\text{in}}))$ is used to calculate the weights, it takes $\tilde{F}_{\text{in}}$ as input. A global average pooling (GAP) layer is used to compress the dimension to $N \times 1$. W_δ is a fully connected layer of dimension $N \times K$, mapping the input to a weight of dimension K . $\delta(\cdot)$ satisfies

$$\sum_{k=1}^K \delta_k(\tilde{F}_{\text{in}}) = 1, 0 \leq \delta_k(\tilde{F}_{\text{in}}) \leq 1. \quad (10)$$

Finally, the output feature F_{out} of the whole MVCT is obtained after the normalization layer.

D. Variable Correlation Self-Distillation

The proposed overall framework performs internal inference on historical series features and uses these features to make predictions for future series. The features in the second stage are more complete and have a wider time window (historical series + future series), so the variable relationship in the second stage is more reasonable. We distill the variable correlation in the second stage into the first stage. The calculation is shown in Fig. 4, the variable correlation matrices in the two networks are averaged to obtain $\{S_I, S_E\} \in \mathbb{R}^{N \times N}$, and the Softmax function is used to calculate the variable correlation score

$$\tilde{S}_I(i, j) = \frac{\exp(S_I(i, j)/\tau_I)}{\sum_{j=1}^N \exp(S_I(i, j)/\tau_I)} \quad (11)$$

$$\tilde{S}_E(i, j) = \frac{\exp(S_E(i, j)/\tau_E)}{\sum_{j=1}^N \exp(S_E(i, j)/\tau_E)}. \quad (12)$$

The Kullback–Leibler (KL) divergence loss [37] is then used to compute the self-distillation loss between the two

$$\begin{aligned} L_{\text{SD}} &= \tau_I \tau_E \text{KL}(S_I \| S_E) \\ &= \tau_I \tau_E \sum_{j=1}^N S_E(i, j) \cdot \log \frac{S_E(i, j)}{S_I(i, j)} \end{aligned} \quad (13)$$

where τ_I and τ_E are two temperature coefficients used to control the smoothness of the two distributions. Since the internal inference process and external inference process of the proposed method are symmetric, we set the ratio between τ_I and τ_E to 1:1. According to [37], the specific value is 1.0.

E. Loss

The loss of the overall framework consists of the interinference network loss L_{Rect} , the exter-inference network loss L_{Pred} , and the variable correlation self-distillation loss L_{SD} . L_{Rect} is computed from the reconstructed historical time series X_{Rect} and the original historical time series $\tilde{X}$

$$L_{\text{Rect}} = \|\tilde{X} - X_{\text{Rect}}\|. \quad (14)$$

L_{Pred} is computed from the predicted future series Y_{Pred} and the true future series $\tilde{Y}$

$$L_{\text{Pred}} = \|\tilde{Y} - Y_{\text{Pred}}\|. \quad (15)$$

The overall loss function of the network is as follows:

$$L_{\text{All}} = L_{\text{Rect}} + L_{\text{Pred}} + L_{\text{SD}}. \quad (16)$$

V. EXPERIMENTAL SETUP

A. Datasets

We verified the proposed method on both long-term forecasting task and short-term forecasting task, using a total of eight datasets. ETT [38] dataset, ECL dataset [38], Exchange dataset [39], and Weather dataset [38] are used in the long-term forecasting task. The ECL dataset collected the electricity consumption of 321 clients between 2010 and 2014, with a 15 min interval. The Exchange dataset records daily exchange rate data for eight countries. The Weather dataset collected 21 weather factors with a time interval of 10 min. The ETT dataset collects transformer load and temperature over a period of two years, with six power load variables and one temperature variable. The ETT dataset contains four subsets: ETTh1, ETTh2, ETTm1, and ETTm2. The input series length of the long-term forecasting task is set to 96 and the prediction series length is set to {96, 192, 336, 720}.

The short-term forecasting uses the PEMS dataset [40], which collects a large amount of sensor data from California's highway system to record the transportation network. The time interval is 5 min. We used three subsets of PEMS (PEMS03, PEMS04, and PEMS08). The input series length of the short-term forecasting task is set to 96 and the prediction series length is set to {12, 24, 48, 96}.

B. Training Details

All datasets are divided into training set, validation set, and test set in a ratio of 70%, 10%, and 20%. The time series data are normalized before entering the network. The whole training process lasted for 15 epochs. The learning rate adopts the decay learning rate strategy, the initial learning rate is set to $5 \times 10e - 4$. The learning rate decays to half of before if the loss does not decrease in three epochs. The optimizer uses ADAM,

and the overall network is implemented using the PyTorch framework and trained on an NVIDIA RTX 3090Ti 24GB GPU. The batch size is set to 32.

C. Evaluation Metrics

The common evaluation metrics mean square error (mse) and mean absolute error (MAE) are adopted in this article. For short-term forecasting task, [40] also suggests using the mean absolute percentage error (MAPE) metric, and we also report the MAPE for the PEMS dataset.

V. EXPERIMENTAL RESULT AND DISCUSSION

A. Long-Term Forecasting Result

The experimental results in the long-term forecasting task are shown in Table I. The proposed method is compared with the latest time series forecasting methods, including iTransformer [28], TimeMixer [24], MICN [23], DLinear [34], PatchTST [35], Crossformer [27], Pyraformer [36], and Autoformer [5]. We retrain these comparison methods and add the same random data missing in the training process for a fair comparison.

We set five data missing rates {0.1, 0.2, 0.3, 0.4, 0.5} for the experiment. The proposed method has the top ranked results in most of the metrics. Our method obtains a total of 67 first ranked metrics and 10 s ranked metrics in the Table I. Other methods that perform well include MICN, TimeMixer, PatchTST, and Crossformer. In addition, the performance of the proposed method is close to other methods at low data missing rate, and has a large lead at high data missing rate. For example, in the results of ETTm2 dataset, the proposed method only achieved the second place at the missing rate of 0.1, but MICN achieved the first place at the missing rate of 0.1. However, when the missing rate is greater than 0.2, the proposed method obtains all the first places and has a large lead. This similar phenomenon is also observed for other datasets.

The visualized prediction results in Fig. 5. Fig. 5 shows a prediction case in the ECL dataset with an input series length of 96 and an output series of 96. The prediction series of the proposed method is closer to the ground true series. Moreover, the proposed method is robust to imperfect reconstruction features in the first stage and does not overfit the reconstruction features. In the prediction series interval [125, 150] in Fig. 5, a curve with a lower peak value appears. The prediction results of other methods are affected by the missing data and directly use the previous temporal regularity (overfitting), and a higher or lower prediction peak value appears between [125, 150]. However, the proposed method is not affected by imperfect reconstructed features and the overfitting temporal regularity, and gives predictions close to the true results.

B. Short-Term Forecasting Result

The experiment results for the short-term forecasting are presented in Table II. The proposed method obtains 37 first ranked metrics and 9 s ranked metrics. In addition, most of the method performance decline with the increase of the missing rate in the short-term forecasting task. In Crossformer and Pyraformer, the

TABLE I
EXPERIMENT RESULTS OF THE PROPOSED METHOD IN LONG-TERM FORECASTING TASK

Datasets	Methods	Ours		iTransformer [28]		TimeMixer [24]		MICN [23]		DLinear [34]		PatchTST [35]		Crossformer [27]		Pyraformer [36]		Autoformer [5]	
		MR	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
ECL	0.1	0.181	0.280	0.185	0.284	0.192	0.290	0.192	0.302	0.236	0.328	0.203	0.298	0.189	0.294	0.284	0.370	0.226	0.336
	0.2	0.187	0.290	0.192	0.296	0.197	0.296	0.196	0.305	0.242	0.337	0.207	0.304	0.193	0.298	0.285	0.371	0.226	0.336
	0.3	0.195	0.300	0.203	0.309	0.202	0.303	0.202	0.311	0.268	0.365	0.213	0.311	0.199	0.304	0.286	0.371	0.234	0.341
	0.4	0.205	0.312	0.217	0.325	0.210	0.311	0.211	0.319	0.313	0.410	0.223	0.322	0.207	0.312	0.287	0.372	0.249	0.351
	0.5	0.220	0.327	0.237	0.343	0.224	0.325	0.223	0.330	0.379	0.466	0.244	0.342	0.218	0.323	0.289	0.374	0.279	0.370
	AVG	0.198	0.302	0.207	0.312	0.205	0.305	0.205	0.313	0.288	0.381	0.218	0.315	0.201	0.306	0.286	0.372	0.243	0.347
Weather	0.1	0.261	0.289	0.261	0.291	0.251	0.290	0.273	0.320	0.267	0.309	0.257	0.291	0.262	0.319	0.298	0.372	0.454	0.479
	0.2	0.259	0.295	0.261	0.303	0.257	0.306	0.279	0.333	0.273	0.327	0.259	0.305	0.264	0.322	0.290	0.350	0.479	0.501
	0.3	0.263	0.307	0.270	0.324	0.269	0.326	0.294	0.353	0.288	0.351	0.268	0.326	0.269	0.329	0.284	0.346	0.516	0.529
	0.4	0.275	0.328	0.289	0.351	0.289	0.351	0.317	0.379	0.312	0.381	0.287	0.353	0.277	0.339	0.281	0.348	0.563	0.560
	0.5	0.297	0.356	0.318	0.384	0.316	0.380	0.349	0.409	0.345	0.414	0.318	0.388	0.289	0.354	0.286	0.356	0.621	0.594
	AVG	0.271	0.315	0.280	0.331	0.276	0.331	0.302	0.359	0.297	0.356	0.278	0.333	0.272	0.332	0.288	0.351	0.527	0.532
ETTh1	0.1	0.461	0.448	0.485	0.468	0.447	0.454	0.478	0.482	0.463	0.450	0.458	0.451	0.532	0.524	0.898	0.741	0.509	0.491
	0.2	0.461	0.455	0.490	0.480	0.469	0.477	0.487	0.490	0.466	0.459	0.468	0.468	0.555	0.541	0.903	0.745	0.513	0.499
	0.3	0.472	0.468	0.507	0.499	0.497	0.500	0.504	0.507	0.488	0.481	0.493	0.493	0.584	0.562	0.912	0.750	0.525	0.515
	0.4	0.494	0.489	0.535	0.525	0.530	0.525	0.529	0.529	0.526	0.512	0.530	0.523	0.623	0.587	0.925	0.757	0.550	0.538
	0.5	0.531	0.517	0.578	0.558	0.568	0.550	0.561	0.554	0.580	0.549	0.582	0.558	0.671	0.615	0.943	0.766	0.584	0.565
	AVG	0.484	0.476	0.519	0.506	0.502	0.501	0.512	0.513	0.504	0.490	0.506	0.499	0.593	0.566	0.916	0.752	0.536	0.522
ETTh2	0.1	0.363	0.399	0.395	0.427	0.424	0.444	0.452	0.453	0.461	0.461	0.384	0.417	0.896	0.668	2.636	1.381	0.440	0.471
	0.2	0.371	0.410	0.465	0.473	0.534	0.508	0.533	0.501	0.554	0.517	0.441	0.461	0.968	0.693	2.592	1.368	0.514	0.523
	0.3	0.398	0.434	0.585	0.541	0.682	0.581	0.666	0.574	0.697	0.593	0.547	0.525	1.056	0.725	2.542	1.353	0.644	0.596
	0.4	0.459	0.476	0.759	0.626	0.857	0.659	0.853	0.664	0.892	0.684	0.707	0.607	1.171	0.767	2.488	1.336	0.832	0.684
	0.5	0.574	0.542	0.987	0.725	1.062	0.741	1.090	0.764	1.136	0.783	0.930	0.706	1.313	0.819	2.433	1.317	1.077	0.783
	AVG	0.433	0.452	0.638	0.559	0.712	0.587	0.719	0.591	0.748	0.608	0.602	0.543	1.081	0.734	2.539	1.351	0.702	0.611
ETTm1	0.1	0.411	0.412	0.412	0.419	0.394	0.415	0.382	0.406	0.417	0.411	0.397	0.411	0.570	0.545	0.770	0.638	0.541	0.497
	0.2	0.415	0.421	0.418	0.428	0.405	0.426	0.392	0.420	0.425	0.424	0.415	0.432	0.585	0.556	0.756	0.634	0.539	0.504
	0.3	0.427	0.436	0.434	0.445	0.421	0.442	0.415	0.443	0.449	0.449	0.441	0.458	0.600	0.569	0.744	0.632	0.559	0.525
	0.4	0.449	0.457	0.462	0.472	0.457	0.475	0.452	0.474	0.491	0.483	0.477	0.489	0.621	0.584	0.735	0.632	0.595	0.552
	0.5	0.486	0.486	0.505	0.507	0.528	0.528	0.501	0.509	0.551	0.524	0.526	0.526	0.644	0.600	0.734	0.638	0.646	0.583
	AVG	0.438	0.443	0.446	0.454	0.441	0.457	0.428	0.450	0.467	0.458	0.451	0.463	0.576	0.532	0.748	0.635	0.576	0.532
ETTm2	0.1	0.309	0.365	0.323	0.378	0.391	0.414	0.313	0.369	0.331	0.385	0.303	0.360	1.277	0.719	1.590	0.921	0.338	0.396
	0.2	0.358	0.404	0.399	0.439	0.495	0.470	0.378	0.420	0.416	0.448	0.353	0.408	1.288	0.723	1.543	0.905	0.408	0.457
	0.3	0.441	0.456	0.526	0.518	0.580	0.515	0.503	0.501	0.558	0.535	0.451	0.477	1.296	0.728	1.491	0.889	0.524	0.532
	0.4	0.570	0.526	0.708	0.611	0.689	0.576	0.689	0.603	0.757	0.635	0.606	0.566	1.302	0.735	1.433	0.873	0.695	0.625
	0.5	0.755	0.615	0.944	0.716	0.855	0.667	0.936	0.716	1.012	0.746	0.825	0.671	1.312	0.747	1.379	0.859	0.926	0.730
	AVG	0.487	0.474	0.580	0.532	0.602	0.528	0.564	0.522	0.615	0.550	0.507	0.496	1.295	0.730	1.487	0.890	0.578	0.548
Exchange	0.1	0.288	0.380	0.347	0.427	0.653	0.493	0.342	0.422	0.342	0.417	0.328	0.405	0.903	0.729	1.292	0.909	0.381	0.452
	0.2	0.304	0.398	0.427	0.498	0.713	0.544	0.358	0.434	0.365	0.430	0.435	0.475	0.910	0.728	1.295	0.911	0.387	0.483
	0.3	0.387	0.444	0.574	0.593	0.643	0.583	0.447	0.506	0.460	0.501	0.555	0.558	0.931	0.735	1.303	0.915	0.461	0.548
	0.4	0.533	0.519	0.782	0.701	0.716	0.664	0.605	0.611	0.620	0.607	0.693	0.649	0.977	0.756	1.322	0.923	0.607	0.639
	0.5	0.732	0.614	1.042	0.816	0.956	0.785	0.831	0.732	0.855	0.731	0.890	0.754	1.044	0.791	1.355	0.937	0.830	0.751
	AVG	0.449	0.471	0.634	0.607	0.736	0.614	0.516	0.541	0.528	0.537	0.580	0.568	0.953	0.748	1.314	0.919	0.533	0.575

The input series length is 96, and the result is the average of the four output series {96, 192, 336, 720}. MR refers to the data missing rate. The red number represents the first rank, and the green number represents the second rank.

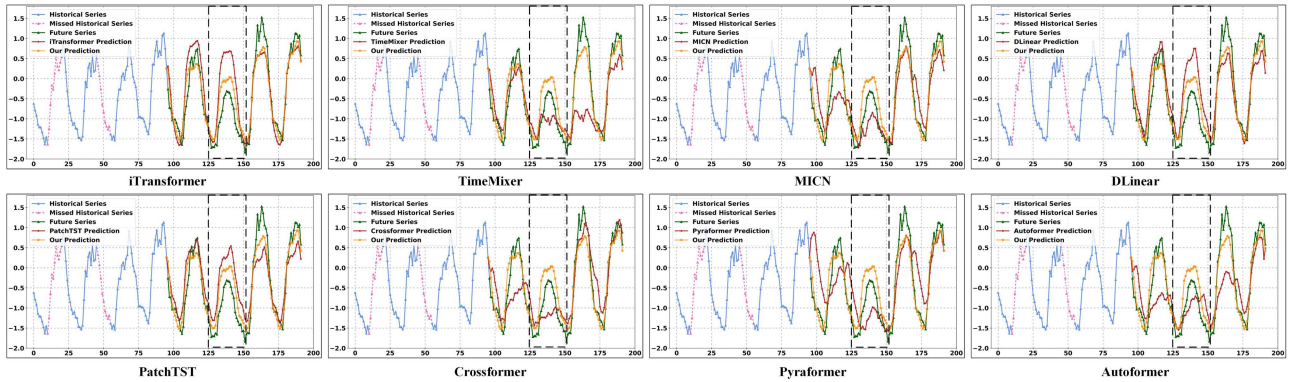


Fig. 5. Visualization of the ECL dataset prediction results. The length of the historical series is 96, and the length of the predicted series is 96.

performance changes little with the increase of missing rate, and Pyraformer performs well in MAE metric of PEMS08 dataset. This indicates that adding random data missing to the training process can make the model robust to a certain extent, even if these models are not specifically designed for missing data.

We also show a visualization result of PEMS03 dataset in Fig. 6. It can be seen that the missing data have a great impact on some models, such as the prediction curve of the DLinear model and the Autoformer model. The prediction curve of the proposed method is very close to the ground true series.

C. Ablation Studies

To explore the effectiveness of the proposed method, we conduct ablation studies on the overall strategy and MVCT.

The overall strategy of the proposed method is the two-stage reconstruction and prediction, and the variable correlation self-distillation. We set up three ablation experiments: without Rect, without SD, and without Rect and SD, the results are presented in the first three rows of Table III. Without Rect refers to removing the reconstruction predictor and loss of the first stage. The mse of the Weather dataset is decreased by 0.035, the mse of the ETTh2 dataset is decreased by 0.206, and the mse of the Exchange dataset is decreased by 0.126. It can be seen that the reconstruction of features has a large impact on the missing series. Without SD refers to removing the variable correlation self-distillation. The mse decreases by only 0.017 for the Weather dataset, 0.013 for the ETTh2 dataset, and 0.008 for the Exchange dataset. This indicates that the self-distillation process has less influence than the reconstruction process, which is in

TABLE II
EXPERIMENT RESULTS OF THE PROPOSED METHOD IN SHORT-TERM FORECASTING TASK

Methods	Datasets	PEMS03							PEMS04							PEMS08						
		MR	0.1	0.2	0.3	0.4	0.5	AVG	0.1	0.2	0.3	0.4	0.5	AVG		0.1	0.2	0.3	0.4	0.5	AVG	
Autoformer [5]	MSE		0.773	0.720	0.683	0.676	0.679	0.706	0.683	0.656	0.627	0.620	0.629	0.643		0.601	0.581	0.596	0.630	0.664	0.614	
	MAE		0.656	0.628	0.612	0.611	0.619	0.625	0.618	0.606	0.597	0.600	0.614	0.607		0.563	0.557	0.569	0.592	0.617	0.580	
	MAPE		3.624	3.839	3.596	3.375	3.167	3.520	3.584	3.341	3.101	2.844	2.664	3.107		3.782	3.653	3.505	3.419	3.228	3.518	
Pyraformer [36]	MSE		0.211	0.207	0.204	0.204	0.209	0.207	0.214	0.212	0.212	0.214	0.220	0.214		0.261	0.263	0.265	0.268	0.276	0.267	
	MAE		0.319	0.316	0.313	0.314	0.321	0.317	0.328	0.326	0.325	0.327	0.336	0.329		0.275	0.276	0.278	0.283	0.292	0.281	
	MAPE		2.880	2.814	2.743	2.662	2.571	2.734	2.481	2.425	2.358	2.280	2.186	2.347		2.310	2.296	2.281	2.274	2.276	2.287	
Crossformer [27]	MSE		0.193	0.192	0.192	0.195	0.205	0.196	0.220	0.221	0.225	0.231	0.243	0.228		0.284	0.282	0.283	0.288	0.302	0.288	
	MAE		0.309	0.306	0.306	0.310	0.321	0.311	0.331	0.332	0.335	0.341	0.353	0.339		0.320	0.319	0.320	0.326	0.342	0.326	
	MAPE		2.642	2.608	2.572	2.527	2.479	2.566	2.420	2.419	2.411	2.397	2.356	2.401		2.523	2.491	2.445	2.405	2.357	2.445	
PatchTST [35]	MSE		0.182	0.183	0.188	0.196	0.209	0.192	0.207	0.206	0.209	0.216	0.232	0.214		0.240	0.244	0.252	0.265	0.287	0.258	
	MAE		0.297	0.299	0.304	0.311	0.325	0.307	0.319	0.318	0.322	0.328	0.343	0.326		0.317	0.318	0.324	0.333	0.351	0.329	
	MAPE		2.109	2.135	2.173	2.200	2.192	2.162	2.111	2.087	2.107	2.121	2.120	2.110		2.226	2.202	2.210	2.210	2.186	2.207	
DLinear [34]	MSE		0.304	0.307	0.329	0.369	0.428	0.348	0.300	0.305	0.330	0.374	0.437	0.349		0.382	0.386	0.413	0.460	0.529	0.434	
	MAE		0.391	0.402	0.428	0.467	0.516	0.441	0.385	0.397	0.426	0.470	0.524	0.440		0.411	0.422	0.451	0.492	0.544	0.464	
	MAPE		2.640	2.407	2.205	1.998	1.819	2.214	2.404	2.210	2.032	1.885	1.735	2.053		2.759	2.528	2.292	2.092	1.904	2.316	
MICN [23]	MSE		0.255	0.262	0.285	0.329	0.391	0.304	0.170	0.172	0.190	0.226	0.285	0.209		0.232	0.237	0.259	0.302	0.369	0.280	
	MAE		0.361	0.373	0.399	0.441	0.492	0.413	0.296	0.296	0.316	0.352	0.406	0.333		0.314	0.317	0.339	0.380	0.436	0.358	
	MAPE		2.571	2.387	2.220	2.087	1.976	2.248	2.191	2.118	2.058	2.017	2.013	2.080		2.512	2.378	2.240	2.119	2.031	2.256	
TimeMixer [24]	MSE		0.200	0.193	0.194	0.201	0.221	0.202	0.203	0.188	0.191	0.203	0.235	0.204		0.240	0.242	0.250	0.265	0.296	0.258	
	MAE		0.313	0.307	0.309	0.315	0.333	0.315	0.316	0.305	0.307	0.318	0.344	0.318		0.326	0.324	0.328	0.337	0.360	0.335	
	MAPE		2.263	2.257	2.257	2.237	2.201	2.243	2.143	2.070	2.058	2.045	2.057	2.075		2.479	2.456	2.417	2.374	2.326	2.411	
iTransformer [28]	MSE		0.179	0.178	0.184	0.190	0.202	0.186	0.194	0.188	0.189	0.193	0.203	0.193		0.221	0.222	0.230	0.243	0.258	0.235	
	MAE		0.290	0.292	0.299	0.308	0.322	0.302	0.301	0.300	0.303	0.310	0.321	0.307		0.309	0.309	0.315	0.324	0.339	0.319	
	MAPE		2.036	2.108	2.166	2.184	2.162	2.131	2.074	2.108	2.126	2.113	2.053	2.095		2.300	2.333	2.356	2.368	2.344	2.341	
Ours	MSE		0.153	0.151	0.155	0.162	0.172	0.159	0.175	0.166	0.167	0.173	0.183	0.173		0.196	0.197	0.208	0.226	0.245	0.214	
	MAE		0.266	0.266	0.272	0.280	0.292	0.275	0.285	0.280	0.283	0.290	0.301	0.288		0.289	0.289	0.295	0.305	0.319	0.299	
	MAPE		1.931	1.976	2.053	2.122	2.156	2.047	1.996	2.000	2.030	2.058	2.056	2.028		2.261	2.255	2.284	2.299	2.307	2.281	

The input series length is 96, and the result is the average of the four output series {12, 24, 48, 96}. MR refers to the data missing rate. The red number represents the first rank, and the green number represents the second rank.

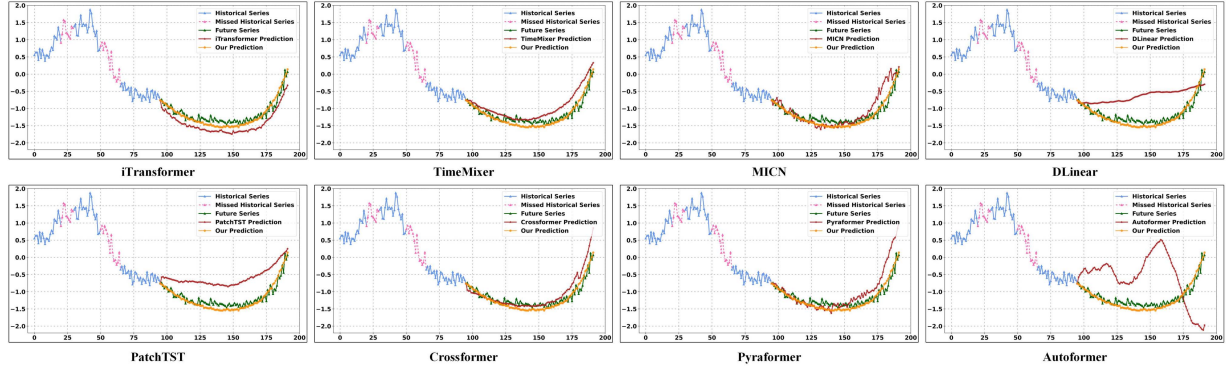


Fig. 6. Visualization of the PEMS03 dataset prediction results. The length of the historical series is 96, and the length of the predicted series is 96.

TABLE III
ABLATION STUDIES ON THE OVERALL STRATEGY AND MVCT

Methods	Weather		ETTh2		Exchange	
	MSE	MAE	MSE	MAE	MSE	MAE
Without Rect	0.298 $\pm$ 0.045	0.346 $\pm$ 0.041	0.639 $\pm$ 0.204	0.566 $\pm$ 0.104	0.575 $\pm$ 0.256	0.574 $\pm$ 0.153
Without SD	0.280 $\pm$ 0.044	0.318 $\pm$ 0.035	0.446 $\pm$ 0.090	0.458 $\pm$ 0.057	0.457 $\pm$ 0.276	0.483 $\pm$ 0.155
Without Rect and SD	0.300 $\pm$ 0.043	0.348 $\pm$ 0.040	0.640 $\pm$ 0.204	0.562 $\pm$ 0.103	0.574 $\pm$ 0.259	0.574 $\pm$ 0.155
Without cosine similarity	0.276 $\pm$ 0.048	0.320 $\pm$ 0.038	0.449 $\pm$ 0.092	0.461 $\pm$ 0.057	0.461 $\pm$ 0.344	0.475 $\pm$ 0.174
Without single-variate FFN	0.272 $\pm$ 0.048	0.315 $\pm$ 0.040	0.436 $\pm$ 0.069	0.453 $\pm$ 0.045	0.453 $\pm$ 0.299	0.458 $\pm$ 0.157
Transformer	0.280 $\pm$ 0.045	0.324 $\pm$ 0.036	0.451 $\pm$ 0.092	0.461 $\pm$ 0.059	0.471 $\pm$ 0.352	0.486 $\pm$ 0.178
Ours	0.263 $\pm$ 0.049	0.313 $\pm$ 0.040	0.433 $\pm$ 0.078	0.452 $\pm$ 0.052	0.449 $\pm$ 0.240	0.471 $\pm$ 0.150

line with the reality, since the first stage of the internal inference and reconstruction process directly affects the missing data. This indicates that the self-distillation process is less influential than the reconstruction process. It is in line with the actual situation, because the inter-inference and reconstruction process in the first stage has a more direct impact on the missing data. Without Rect and SD refers to the deletion of both strategies. The mse of ETTh2 dataset decreases by 0.207 and the MAE decreases by 0.110. Both strategies have positive effects on performance improvement.

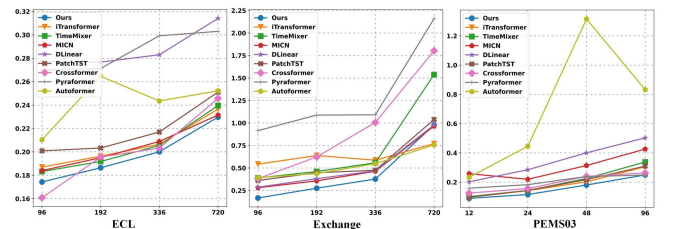


Fig. 7. Effect of different series lengths on performance (mse).

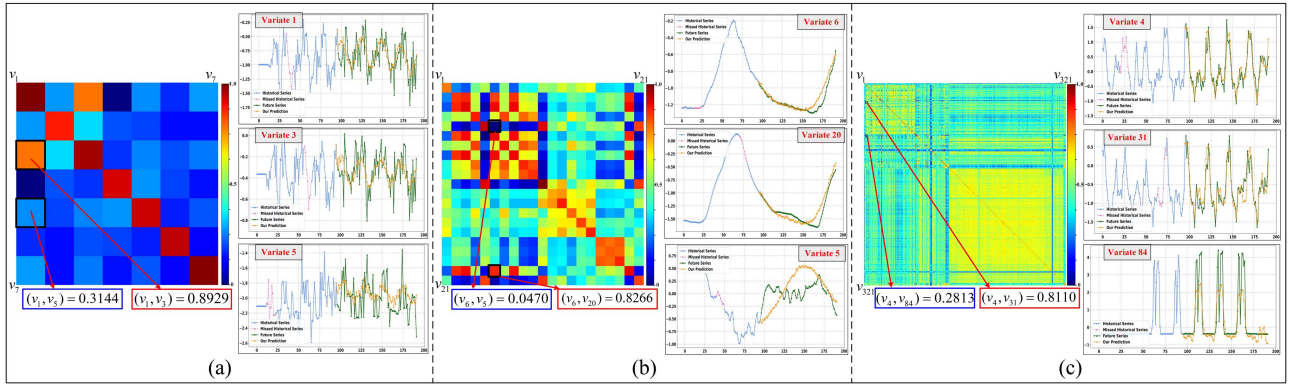


Fig. 8. Visualized variable correlation heatmap. (a) ETTh2 dataset, with seven variables. (b) Weather dataset, with 21 variables. (c) ECL dataset, with 321 variables. Each example shows a set of highly correlated variables and a set of low-correlated variables.

For the ablation studies of MVCT, we also set up three experiments: without cosine similarity, without single-variate FNN, and the traditional Transformer, the results are shown in the fourth to sixth rows of Table III. In without cosine similarity, the mse and MAE of the ETTh2 dataset are 0.449 and 0.461, and the mse and MAE of the Exchange are 0.461 and 0.475. However, in without single-variate FNN, the mse is 0.436 and MAE is 0.453 for the ETTh2 dataset, and the mse is 0.453 and MAE is 0.458 for the Exchange dataset. It can be seen that both of cosine similarity and single-variate FNN have a positive impact on performance. The effect of without single-variate FNN is smaller and the result is closer to the final network. In addition, there are some differences of ablation studies results across datasets, and it can be seen that the results of this three studies are close in Weather dataset.

Fig. 7 shows the results for different prediction series lengths for the three datasets. It can be seen that the performance of most methods decreases as the prediction series length increases. The proposed method has better performance than other method. As the length of the prediction series increases, the lead is more obvious.

D. Analysis of Feature Correlation and Effectiveness

The proposed method uses cosine similarity to construct the Transformer, and uses the self-distillation method to strengthen the correlation. To verify the constructed feature correlation is correct, the visualized feature correlation map of MVCT is shown in Fig. 8. The heatmaps on the left show the correlation between any two variables. We normalize the correlation to be $[0, 1]$. It can be seen that the proposed MVCT can accurately capture similar variables. The variable correlation attention in MVCT adopts a learnable way, which can fit a good variable relationship from a large number of time series data, and can deal with the nonlinear dependence between variables. There are similar curves between highly correlated variables, while the curves between low correlated variables are not similar. Moreover, MVCT can achieve accurate capture even if the number of variables in the dataset is large. For example, the ECL dataset has a total of 321 variables, and we show the curves for

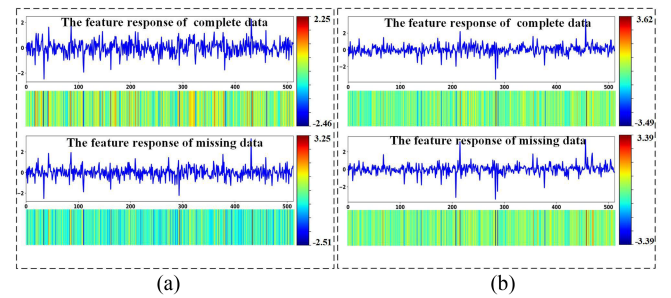


Fig. 9. Comparison of the missing data and complete data features. (a) Comparison of the two features in iTransformer. (b) Comparison of the two features in the proposed method.

its variate 4 and variate 31 in Fig. 8(c). It can be seen that the curves for these two variables are very similar, and the variable correlation calculated by MVCT is 0.811.

In addition, in order to demonstrate the robustness of MVCT. Features (a 512-D vector) are collected from the MVCT output, and the features of missing and complete data are compared. The feature response curves and feature intensities heatmaps are presented in Fig. 9. We compare the iTransformer method [see Fig. 9(a)] with the proposed method [see Fig. 9(b)]. It can be seen that the features of the proposed method in the case of missing data are closer to the case of complete data, which indicates that the proposed method has better robustness to missing data. To verify the effectiveness of the reconstructed features, we show the initial time series feature F_{in} and the complementary feature $\tilde{V}$ in Fig. 10. It can be seen that the two are similar overall, but they are significantly different in the missing information location, and the missing information has a direct impact on F_{in} . We also quantitatively analyze the results using only F_{in} , whose mse is 0.621 and MAE is 0.537 on ETTh2 dataset, which are much lower than the final network.

E. Analysis of Computation Cost

Table IV shows the floating point operations (FLOPs), the number of parameters (Params), and the prediction time. Three conclusions can be drawn from the analysis of the computation

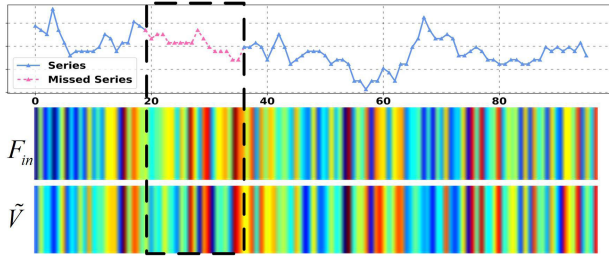


Fig. 10. Comparison of original feature and complementary feature.

TABLE IV
COMPUTATION COST ANALYSIS

Datasets	Methods	FLOPs (G)	Params (M)	Time (s)
ETTh2	DLinear [34]	0.0001	0.0186	0.56
	MICN [23]	0.0091	0.1092	1.38
	Crossformer [27]	0.0742	1.6647	0.55
	Pyraformer [36]	0.0261	0.4770	1.34
	Autoformer [5]	0.0424	0.3730	1.42
	Ours	0.0049	0.4480	1.31
Weather	Ours	0.0420	1.6831	1.34
ECL	Ours	2.1153	6.5118	1.35

cost. First, the proposed method is compared with several existing methods on the ETTh2 dataset. It can be seen that although the proposed method is a two-stage framework, the FLOPs and Params are not the highest and within an acceptable range. The test time of the proposed method is only 1.31 s, which is close to most of the methods. Second, Table IV also shows the results of the proposed method on the Weather dataset and ECL datasets. The Params and FLOPs also increase due to the greatly increased number of variables in these two datasets. However, the test time does not increase greatly, which indicates that the current hardware resources can fully meet the computation cost of the proposed model. Even though the number of variables and model parameters increased, the test time remained at a low level. Third, from the perspective of the overall computation cost, the model scale and computation of time series forecasting tasks are less, and the demand for hardware platforms is less.

VI. CONCLUSION

In this article, we propose a multivariate correlation self-distillation Transformer framework to solve the incomplete time series forecasting. The proposed method is a two-stage framework, where the first stage focuses on inter-inference of historical series and the second stage focuses on exter-inference of future series. The MVCT is designed to capture the complementary information of similar variables, and the self-distillation method strengthens the variable correlation between the two-stage network. We conduct missing data time series forecasting experiment on eight datasets, which cover various fields. The experiment results of long-term and short-term forecasting all show that the method has good prediction effect in different data missing rates (0%–50%). Moreover, the effectiveness and robustness of the method are demonstrated by ablation studies and visualized feature maps.

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